## a Validation: distinct denominators

| Comparison / agreement criterion | Met / tested units | Not met, n |
|---|---|---|
| 8088/hardware: identical vector states | 3,126,917/3,137,000 | 10,083 |
| Corrected/historical: below change-rate bound | 643/690 | 47 |
| Corrected/historical: identical assay metrics | 407/417 | 10 |
| Corrected/historical: unchanged claim verdicts | 19/22 | 3 |
| Corrected/historical: unchanged components | 32/36 | 4 |

All 10,083 hardware divergences were expected

Stable: upper 95% CI of changed-final-memory pair fraction <0.01%
100,000 encounter draws/condition; units named in each row

## b I1: flags remain flags

| Core | Flagged / conditions | Assays differing / tested |
|---|---|---|
| Corrected | 47/690 | 10/417 |
| C2 | 114/690 | 10/417 |

Comparator: historical compatibility core C0

Corrected core (C1): classic; core C2: Zilog NMOS

Unchanged across both comparisons: criterion not met

Stable: upper 95% CI of changed-final-memory pair fraction <0.01%
Flag = stability criterion not met; assays compare with core C0

## c I1-T2: all 22 claim verdicts

| Claim | vs archive | Claim | vs archive |
|---|---|---|---|
| DZ1 | stands | C4b | stands |
| DZ2 | changed | C4c | stands |
| DZ3 | changed | F4c | stands |
| E1 | stands | S1 | stands |
| E2 | stands | S2 | stands |
| E3 | stands | S3 | stands |
| E-L32L | stands | S4 | stands |
| L1 | stands | S5 | stands |
| L2 | changed | X1-1 | stands |
| L3 | stands | X1-2 | stands |
| L | stands | X1-3 | stands |

DZ2 timing; DZ3 transmitter trend; L2 recovery sites
Stands = same as archive; E1 stability remains failed; 19/22 stand

## d Four changed components of 36

| Component | Historical | Corrected core |
|---|---|---|
| DZ2 | not met | met |
| DZ3 | not met | met |
| L2 L16 | not met | n/a |
| L2 L20 | n/a | not met |

n/a = not applicable: no recovered lineage

Component tests summarized in panel a; L = tape bytes

DZ2/DZ3: criteria now met; L2: applicability changes

Historical = published archive; DZ timing/share; L2 sites
Complete 36-component ledger retained in Source Data

## e Assay changes by measurement

| Metric | Differing corrected core | Differing C2 |
|---|---|---|
| Partner-copy fraction, parent A | 0 | 0 |
| Partner-copy fraction, parent B | 6 | 6 |
| Initial byte-match fraction | 0 | 0 |
| Informative partners, generation 1 (n) | 0 | 0 |
| Informative partners, generation 2 (n) | 0 | 0 |
| Cyclic copy offset (bytes) | 0 | 0 |
| Parent-byte fraction preserved, A | 0 | 0 |
| Parent-byte fraction preserved, B | 9 | 9 |
| Offspring ≥75% parent-match fraction | 0 | 0 |
| Generation-2 normalized gain | 0 | 0 |
| Generation-2 conditional gain | 0 | 0 |
| Normalized culture gain | 0 | 0 |
| Generation-2 gain ≥0.3 (yes/no) | 0 | 0 |
| Faithful culture classification (yes/no) | 0 | 0 |

Counts /417 assays vs core C0; rows overlap (not additive)
A/B = parent role; copy = ≥75% immediate-parent byte match

## f Stage-L scoring and snapshot availability

| L bytes | Worlds, n | Scoreable from final data, n | Missing 7,500-update snapshot, worlds |
|---|---|---|---|
| 16 | 10 | 10 | 10 |
| 20 | 10 | 10 | 10 |

All worlds completed 10,000,000 updates

Retained final bytes and journals support scoring

Applied-mutation/collision counts unavailable for these worlds

Stage L: closure-recovery worlds, 10 / tape length
The missing observation is a world snapshot at 7,500 updates